

Windows Authentication Log Analysis

Successful logons, incorrect passwords and unknown users

A personal lab examining events 4624 and 4625

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1 October 2026

Lab environment and test procedure

Objective: connect known actions to recorded events, interpret relevant fields and explain the conclusions.

I set up a Windows virtual machine and generated controlled authentication attempts. I then located the corresponding events in the **Security** log using Event Viewer, comparing accounts, logon types, source addresses and error codes.

Component	Lab configuration
Virtualisation	Hyper-V, SOC-Lab-Windows virtual machine, Generation 2
Operating system	Windows 11 Enterprise Evaluation, English language
VM resources	8 GB RAM, 2 virtual processors, 80 GB dynamically expanding disk
Network	Hyper-V Default Switch; authentication attempts performed locally in the VM
Accounts	LabAdmin: local administrator. LabUser: local standard user.
Auditing	Audit Logon configured for Success and Failure

Procedure

- 1 Preparation.** Installed Windows, created the accounts and took a checkpoint of the initial VM state.
- 2 Authentication tests.** Started a command prompt as LabUser using runas, first with an incorrect password and then with the correct one; checked the identity using whoami.
- 3 Comparison.** Attempted another logon using Utentelnesistente to distinguish an incorrect password from an invalid username.
- 4 Collection.** Filtered events 4624 and 4625, inspected the General/XML views and exported the three selected events as EVTX files.

Scope. These tests use local accounts in a personal VM. The lab documents manual log analysis; it does not include a SIEM platform or an analysis script.

Reading a logon event

The Event ID describes the outcome. Other fields identify the accounts involved and the authentication context.

Field	Question it answers
Event ID	4624: successful authentication. 4625: failed authentication.
Subject	Which account or context requested authentication?
New Logon / Account For Which Logon Failed	Which account was logged on, or attempted the failed logon?
Logon Type	What kind of authentication was requested?
Source Network Address	Which source IP address is recorded in the event?
Status / Sub Status	Which cause of authentication failure is recorded?

Requesting account and target account

In the runas test, **Subject = LabAdmin** because the request originates from that session. **LabUser** is the target account. Reading only Subject would attribute the authentication attempt to the wrong account.

Two observed logon types

Type 2 - Interactive. This is the type recorded in the runas tests. The session allows a program to run under the other account; it does not necessarily represent a new desktop sign-in.

Type 5 - Service. During the initial exploration, I also inspected a 4624 event at 13:43:23: New Logon = SYSTEM and the process was `services.exe`. These fields are consistent with a Windows service starting.

Local source. `127.0.0.1` and `::1` are loopback addresses referring to the same machine. The double colon is part of the IPv6 address `::1`.

The General view provides readable text; XML retains named fields such as `TargetUserName`, `LogonType` and `SubStatus`. This report contains excerpts of values copied from the lab events.

Failed logon, then success

Main case: two authentication attempts for LabUser, approximately 13 seconds apart.

Local time	Event ID	Record ID	Result
13:58:39	4625	31479	Incorrect password
13:58:52	4624	31481	Successful authentication

4625 / RECORD 31479 - TRANSCRIBED EXCERPT

Subject Account Name: LabAdmin
Target Account Name: LabUser
Logon Type: 2
Source Address: ::1
Status: 0xC000006D
Sub Status: 0xC000006A

- 1 Identify the attempt.** LabAdmin requests a logon for LabUser. Type 2 and the loopback source are consistent with the local test.
- 2 Determine the failure cause.** The generic message refers to an unknown username or incorrect password. SubStatus **0xC000006A** specifically indicates an incorrect password.
- 3 Compare the successful logon.** The subsequent 4624 records New Logon = LabUser, the same Subject Logon ID 0xCB260, the same calling process and the same local source.

The successful logon creates a new LabUser session with Logon ID 0xA5974E and **Elevated Token = No**. Subject Logon ID instead identifies the requesting session; the two identifiers have different roles.

Case conclusion. The shared fields and close timestamps support linking these two attempts. The known test procedure confirms an incorrect password followed by the correct one; the logs alone would not establish the operator's intent.

Times: 1 October 2026, UTC+02:00. XML timestamps: 11:58:39.8690601Z and 11:58:52.8747152Z. The earlier accidental successful logon at 13:58:09 is excluded from the main case.

Unknown username: comparing causes

A second 4625 event shows why the substatus is more informative than the generic failure message.

At **14:29:46**, I attempted to start a command prompt as **Utentelnesistente**, a deliberately nonexistent local account. The command returned error 1326: the username or password is incorrect. I then located the corresponding event in the Security log.

4625 / RECORD 31496 - TRANSCRIBED EXCERPT

Event ID:

4625

EventRecordID:

31496

Subject Account Name:

LabAdmin

Target Account Name:

UtenteInesistente

Logon Type:

2

Source Address:

::1

Status:

0xC000006D

Sub Status:

0xC0000064

Comparison with the incorrect password

Target account	Status	Sub Status	Interpretation
LabUser	0xC000006D	0xC000006A	Incorrect password
Utentelnesistente	0xC000006D	0xC0000064	Unknown or incorrect username

Both failures share the same Status and general message. Sub Status distinguishes their causes. **NULL SID**, present in both events, does not by itself distinguish an unknown username from an incorrect password.

Comparison conclusion. The second event records a local authentication attempt using an invalid username. In this test, the account is deliberately nonexistent; an isolated event would not establish that an attack occurred.

XML timestamp: 12:29:46.1872612Z. This attempt is separate from the main case and is retained as a diagnostic comparison. Account names are preserved exactly as recorded.

Conclusions, evidence and references

Observed results, interpretation limits and retained evidence.

What I verified

I linked known actions to Windows events, distinguished requesting and target accounts, examined logon types and source addresses, compared two failure causes and reconstructed a failure-success sequence.

Facts and interpretation. Logs record authentication outcomes and context. The fact that these are intentional, controlled tests comes from the known procedure, not the Event ID itself. A failure followed by a success is not sufficient evidence of compromise.

Evidence retained in the lab

EVTX file	Event ID / Record ID
01-accesso-fallito.evtx	4625 / 31479
02-accesso-riuscito.evtx	4624 / 31481
03-utente-inesistente.evtx	4625 / 31496

The three events were exported from the VM. LocaleMetaData accompanies the EVTX files with event descriptions in the selected language. **This report is based on text and XML copied from the events**; the original EVTX files remain in the VM and are not included in the public repository. Original filenames are retained above.

Investigation workflow

I performed the authentication tests in the VM, inspected the events in Event Viewer and verified the recorded outcomes. The investigation focused on manual analysis of the Security log.

Microsoft documentation

- Event 4624 - Successful logon
- Event 4625 - Failed logon and error codes
- Audit Logon - Recording authentication attempts
- LocaleMetaData - Descriptions of exported events